

LEVEL 14 — Splitting Arrays in NumPy

Easy line-by-line examples with outputs

1. np.split()

Example:

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50, 60])

result = np.split(arr, 3)

print(result)
```

Output:

```
[array([10, 20]), array([30, 40]), array([50, 60])]
```

Line-by-line explanation:

Line 1: Import NumPy.

Line 3: Create an array containing 6 values.

Line 5: np.split(arr, 3) splits the array into 3 equal parts.

Line 7: Print the three parts.

Important: np.split() normally requires equal-sized parts.

2. np.array_split()

Example:

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

result = np.array_split(arr, 3)

print(result)
```

Output:

```
[array([10, 20]), array([30, 40]), array([50])]
```

Line-by-line explanation:

Line 1: Import NumPy.

Line 3: Create an array containing 5 values.

Line 5: array_split(arr, 3) tries to create 3 parts.

Line 7: Print the parts.

Important: array_split() can split an array into unequal parts when necessary.

3. np.hsplit()

Example:

```
import numpy as np

arr = np.array([[10, 20, 30, 40],
               [50, 60, 70, 80]])

result = np.hsplit(arr, 2)

print(result)
```

Output:

```
[array([[10, 20],
       [50, 60]]),
 array([[30, 40],
       [70, 80]])]
```

Line-by-line explanation:

Line 1: Import NumPy.

Line 3–4: Create a 2D array with 2 rows and 4 columns.

Line 6: `hsplit()` means horizontal split.

It splits the columns into 2 groups.

Line 8: Print both column groups.

4. np.vsplit()

Example:

```
import numpy as np

arr = np.array([[10, 20],
               [30, 40],
               [50, 60],
               [70, 80]])

result = np.vsplit(arr, 2)

print(result)
```

Output:

```
[array([[10, 20],
       [30, 40]]),
 array([[50, 60],
       [70, 80]])]
```

Line-by-line explanation:

Line 1: Import NumPy.

Line 3–6: Create a 2D array with 4 rows.

Line 8: `vsplit()` means vertical split.

It splits the rows into 2 groups.

Line 10: Print both row groups.

5. np.dsplit()

Example:

```
import numpy as np
```

```
arr = np.array([[10, 20],
                [30, 40]],
                [[50, 60],
                [70, 80]])

result = np.dsplit(arr, 2)

print(result)
```

Output:

```
[array([[10],
        [30]],
       [[50],
        [70]]),
 array([[20],
        [40]],
       [[60],
        [80]])]
```

Line-by-line explanation:

Line 1: Import NumPy.

Line 3–6: Create a 3D array.

Line 8: `dsplit()` splits along the third dimension (depth).

Here, the last dimension has size 2, so it is split into 2 parts.

Line 10: Print the two depth parts.

Quick Difference

np.split() → Equal-sized splits.

np.array_split() → Allows unequal-sized splits.

np.hsplit() → Splits columns horizontally.

np.vsplit() → Splits rows vertically.

np.dsplit() → Splits the third/depth dimension of a 3D array.

Easy memory trick: H = Horizontal/columns, V = Vertical/rows, D = Depth/3D.